

Sentiment Classification of Amazon Reviews with Recurrent Neural Networks

Comparing RNN, LSTM, and GRU Architectures, and Studying Class Imbalance and Cross-Dataset Generalization to IMDB

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1. Part One: Comparing RNN, LSTM, and GRU

The goal is to train a sentiment classification system for Amazon product review data using recurrent neural networks. The target dataset is a subset of the Amazon Review Data (2018) 5-core dataset, restricted to the “Industrial and Scientific” category. From this dataset, the reviewText field is used as the input text of each review, and the overall field — a numeric rating from 1 to 5 — is used as its label.

In order to feed text data into the networks, it must first be converted into numbers. This requires the input strings to be tokenized first. Tokenization is performed with spaCy, using one of its built-in vocabularies. For words that do not appear in the vocabulary, the <UNK> token is used.

Next, the performance of RNN, LSTM, and GRU networks with different numbers of layers is evaluated. The accuracy and loss curves for the LSTM network with 1, 2, and 3 layers are shown in Figures 1, 2, and 3. Results for the RNN are shown in Figures 4, 5, and 6, and results for the GRU are shown in Figures 7, 8, and 9.

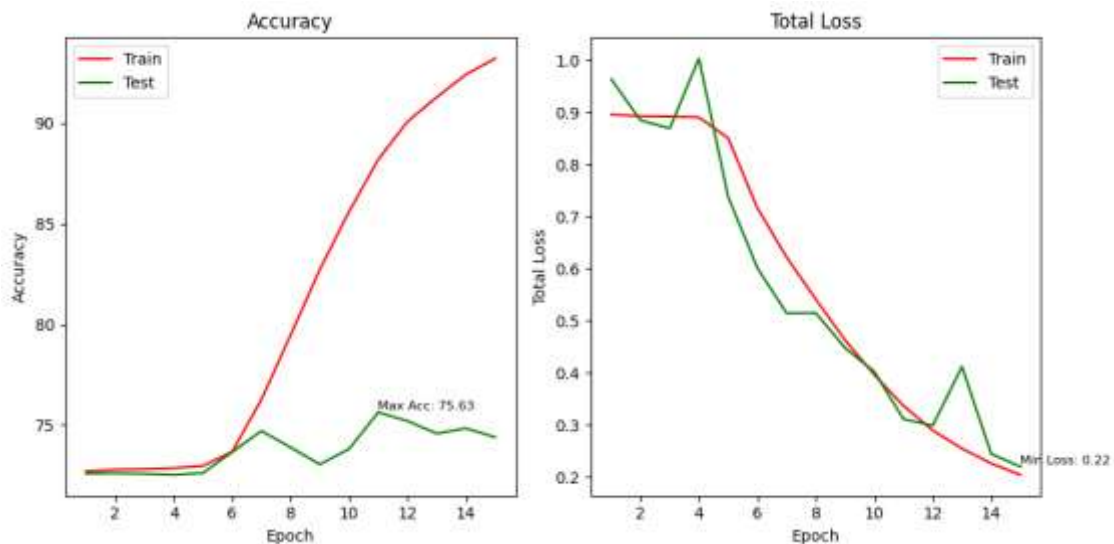


Figure 1: Accuracy and loss curves for the 1-layer LSTM network

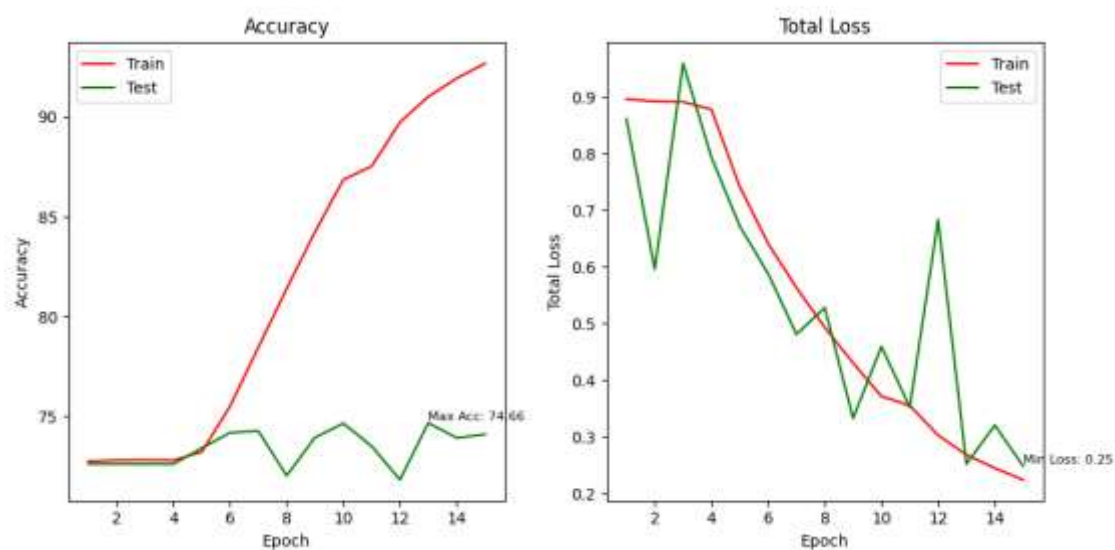


Figure 2: Accuracy and loss curves for the 2-layer LSTM network

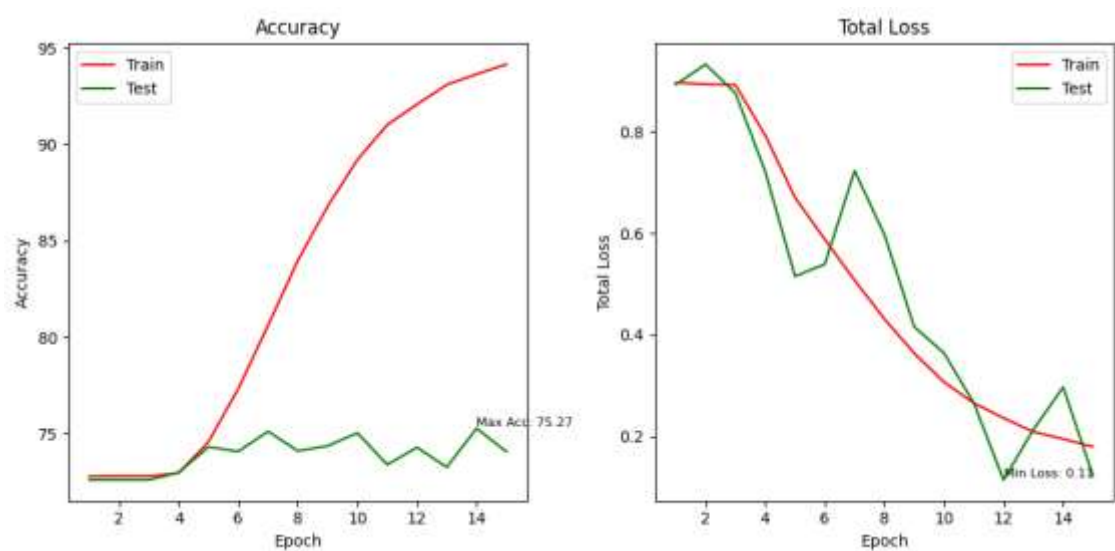


Figure 3: Accuracy and loss curves for the 3-layer LSTM network

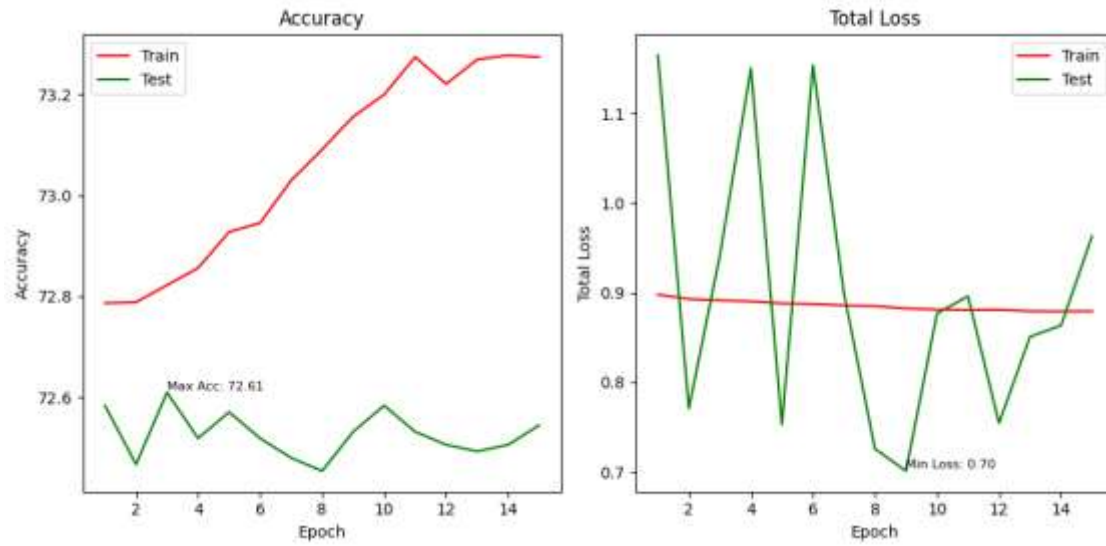


Figure 4: Accuracy and loss curves for the 1-layer RNN network

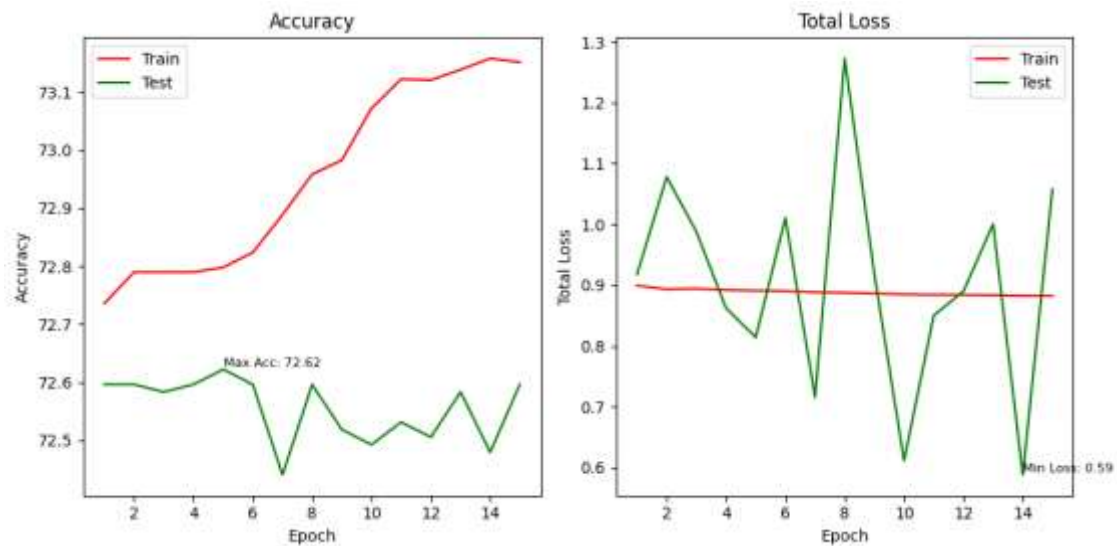


Figure 5: Accuracy and loss curves for the 2-layer RNN network

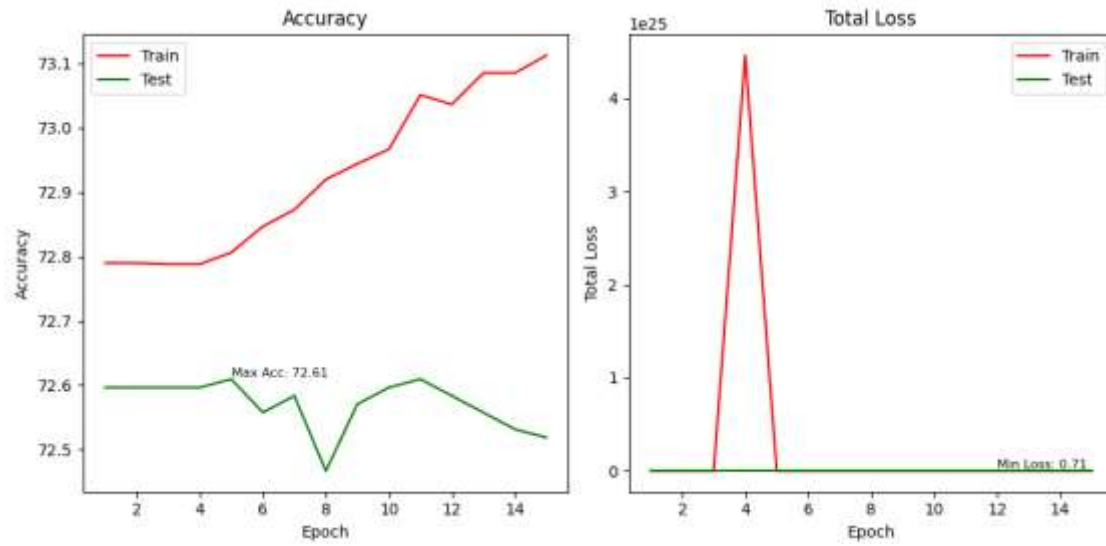


Figure 6: Accuracy and loss curves for the 3-layer RNN network

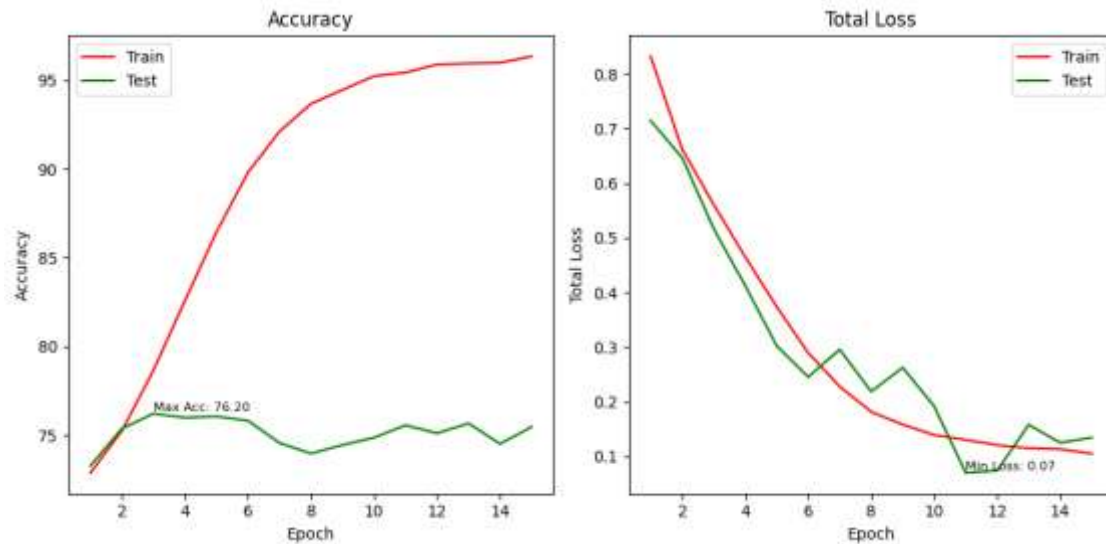


Figure 7: Accuracy and loss curves for the 1-layer GRU network

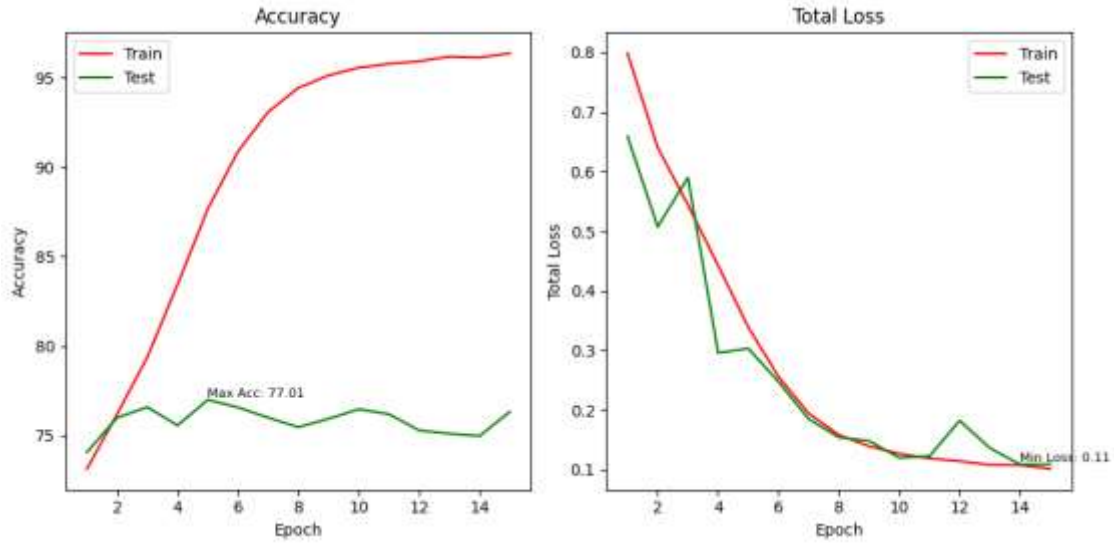


Figure 8: Accuracy and loss curves for the 2-layer GRU network

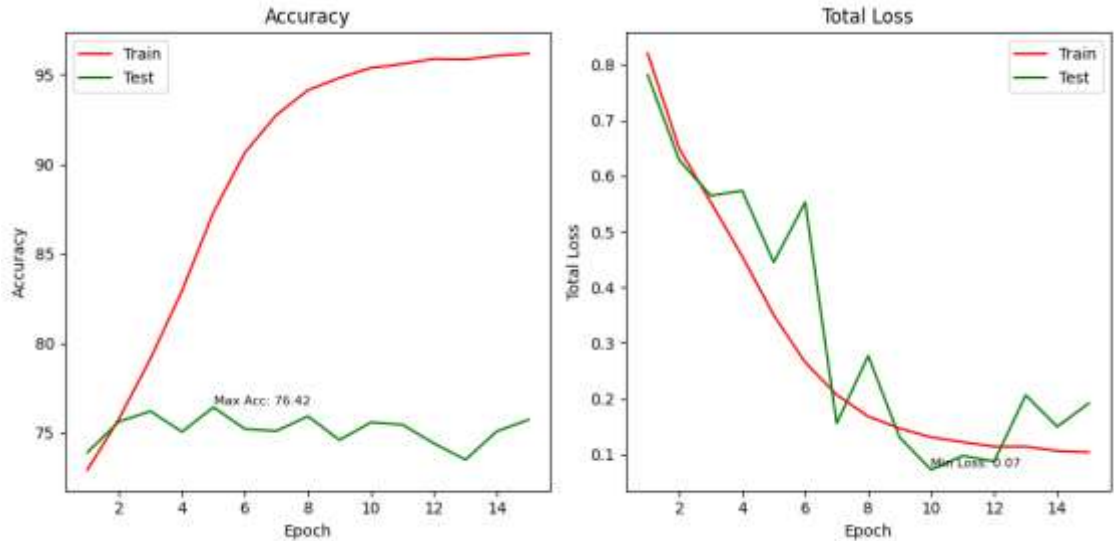


Figure 9: Accuracy and loss curves for the 3-layer GRU network

As shown, the plain RNN is unable to learn this data well regardless of the number of layers used. In contrast, the LSTM and, in particular, the GRU networks learned the data effectively and were able to reduce the loss on both the training and test sets. However, as the curves show, the models overfit the training data: after the first few epochs, no further improvement in test accuracy is observed. This may be due to a shortage of training data relative to the test/validation data, or to an imbalance across classes, which can bias the model's output toward the class with more available samples.

A comparison of network performance for different numbers of layers, based on the best validation accuracy achieved, is shown in Table 1. For the LSTM, the best performance was obtained with 1 layer, followed by the 3-layer configuration. For the plain RNN, changing the number of layers had essentially

no effect on performance. For the GRU, the 2-layer configuration achieved the best accuracy, and the 3-layer configuration outperformed the 1-layer one.

Overall, the best performance was achieved by the 2-layer GRU network, reaching 77.01% accuracy on the validation data.

Network	Number of Layers	Validation Accuracy (%)
LSTM	1	75.63
LSTM	2	74.66
LSTM	3	75.27
RNN	1	72.61
RNN	2	72.62
RNN	3	72.61
GRU	1	76.20
GRU	2	77.01
GRU	3	76.42

Table 1: Comparison of network performance for different numbers of layers

2. Part Two: Class Removal and Cross-Dataset Generalization

In this part, the architecture that performed best in Part One — the 2-layer GRU network — is selected. Data belonging to classes 2 and 4 are then removed from the training set, and the network is retrained from scratch. As shown, because the network never encounters any class-2 or class-4 samples during training, it cannot return the correct score or label for them; consequently, the test accuracy at every epoch during training is zero. The accuracy and loss curves for the training and test data are shown in Figure 10.

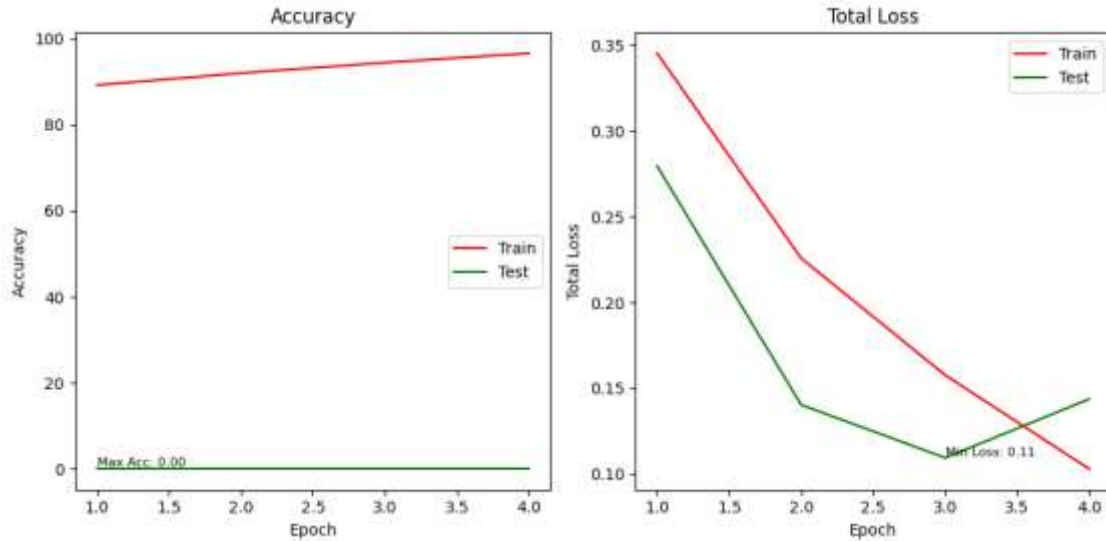


Figure 10: Accuracy and loss curves for a network trained on classes 1, 3, and 5 and tested on classes 2 and 4

The performance of the network trained on the three classes 1, 3, and 5 is also evaluated on the IMDB reviews dataset. This dataset contains IMDB movie reviews labeled as positive or negative. When the data is loaded in batches, positive labels are converted to 1 and negative labels to 0.

To evaluate the network's performance on the IMDB dataset, a batch of this data is selected and passed through the network, and the network's output for each sample is compared with its true label. This procedure was repeated for models saved at different training epochs, and the best performance was obtained with the model saved at epoch 4; the results are shown in Figure 11.

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[0, 0, 1, 1, 0, 1, 0, 1, 1, 0, 1, 0, 0, 1, 1, 1, 1, 1, 0, 0, 0, 1, 0, 1, 1, 1, 0, 0, 0]
[4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 3, 4, 4, 2, 0, 3]
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Figure 11: Model output on IMDB data — top row: ground-truth labels, bottom row: model predictions

As shown, the output for every positive-class sample is class 5 (index 4 in the implementation), which is the expected result. For negative samples, however, some outputs are 4, while others are 0 or 2. This error can again be attributed to the model overfitting due to the class imbalance in the training data.